CREDITSENSE

AI Credit Risk Assessment Platform

A credit score, a probability of default, a clear decision and a plain-English reason, in under two seconds.

Pitch Deck | Prototype overview for banks and risk teams

Built with Python, FastAPI, XGBoost, SHAP, React and Tailwind CSS

The Problem

Slow, manual reviews	Black-box scores	Late risk signals
Loan officers piece together applicant details by hand. Decisions take time and vary from one officer to the next.	Many scoring tools return a number with no reason. Officers cannot explain a rejection and applicants cannot act on it.	Risk managers often see portfolio stress only after defaults rise, when it is too late to adjust lending policy.

Consequence

- Inconsistent approvals, missed good borrowers and avoidable defaults.
- Weak audit trail for regulators who expect explainable and fair lending decisions.
- Concentrated risk that builds quietly across a portfolio and can spread into wider financial stress.

Our Solution

One platform for two audiences: the loan officer at the desk and the risk manager looking at the whole book.

For loan officers	For applicants	For risk managers
Enter applicant details in a guided four-step form. Get a 300 to 850 credit score, probability of default, risk band and an Approve, Manual Review or Reject recommendation.	See the top five factors that drove the result, in plain English, plus two or three concrete steps that would improve the score.	Portfolio dashboard with risk distribution and approval trends, plus a stress test that shows how shocks move loans into higher risk bands.

The tool supports human decisions. It does not replace them.

How It Works

Step	What happens	Output
1. Enter	Officer completes Personal, Financial, Credit and Loan steps with inline validation and sample applicants.	Validated applicant record
2. Derive	Backend computes debt-to-income, loan-to-income, estimated EMI and EMI-to-income automatically.	Model-ready features
3. Score	Calibrated XGBoost or logistic model predicts probability of default. A documented formula maps it to 300 to 850.	Score, risk band, decision
4. Explain	SHAP identifies the top five factors and what-if analysis finds realistic improvements.	Plain-English reasons and advice
5. Record	Assessment is saved to SQLite and feeds the portfolio dashboard and reports.	PDF report and audit history

The Scoring Engine

Personal	Financial	Credit	Loan
Age, employment status, employment length, dependents	Monthly income, monthly expenses, existing debt payments, savings	Credit history length, open credit lines, 30/60/90-day delinquencies, utilization	Loan amount, term in months, loan purpose

Every assessment returns

- **Probability of default** a calibrated value between 0 and 1.
- **Credit score and risk band** 300 to 850 scale; Excellent, Good, Fair, Poor or Very Poor.
- **Decision** Approve, Manual Review or Reject using configurable thresholds.
- **Lending guidance** recommended maximum loan amount and suggested interest rate range.

The score reacts to every input. Higher income lifts it; higher debt or past delinquencies lower it. Monotonicity is covered by automated tests.

Explainable by Design

Factor	Direction	Plain-English explanation
Debt-to-income	Raises risk	Debt payments take 48% of monthly income, well above the safe range of 36%.
Credit utilization	Raises risk	Card utilization is high; keeping it below 30% is a healthier level.
Credit history length	Lowers risk	A longer credit history shows a stable repayment record.
Past delinquencies	Raises risk	Late payments in the last years weigh heavily on the estimate.
Savings balance	Lowers risk	Savings give the applicant a cushion against income disruption.

Illustrative example of the explanation format shown for each applicant.

What would improve this score

- Reduce credit utilization below 30%.
- Lower monthly debt payments or request a smaller loan amount.
- Choose a term that brings EMI to a smaller share of monthly income.

Portfolio Stress Testing

Spot rising risk early, before it becomes a wider financial problem.

Income shock	Interest rate rise	Unemployment rise
Slider from 0 to -30% applied to every borrower.	Slider from 0 to +5% raising EMI burden.	Share of borrowers losing income.

Output

- Expected default rate before and after the shock, with animated chart updates.
- Number of loans migrating into higher risk bands.
- A plain-English summary of the systemic risk impact for the portfolio.

Fairness and Responsible AI

No protected attributes	Disparity monitoring	Human in the loop
Gender, race, religion and marital status are never used as model features.	Approval rates by age band with a simple disparity indicator on the Model Insights screen.	A visible note on every result: the tool supports human decisions and does not replace them.

Transparency commitments

- Only metrics the trained model actually achieves are reported in the product.
- Probabilities are calibrated so a stated risk level matches observed outcomes.
- Known limitations are documented in the README, including the dataset used.

Model and Data

Stage	Approach
Data	Kaggle "Give Me Some Credit" or UCI German Credit; otherwise a documented 15,000-row synthetic generator with realistic correlations.
Preparation	Cleaning, missing-value handling and class-imbalance handling.
Models	Logistic regression baseline compared with XGBoost.
Evaluation	AUC-ROC, precision, recall, F1 and confusion matrix; ROC curve and model comparison table shown in the UI.
Calibration	Probability calibration before the best model is saved with joblib.
Explainability	SHAP values for global feature importance and per-applicant factors.

Final metrics are produced by running the training pipeline and are reported in the product and README.

Product and Technology

Screen	Purpose
Dashboard	KPI cards, risk band and score distributions, approvals over time, stress test panel.
New Assessment	Four-step form, animated score gauge, SHAP bars, suggestions and PDF report download.
Applications	Searchable, sortable, filterable history with a detail drawer; seeded with 200+ sample assessments.
Model Insights	Metrics, ROC curve, comparison, feature importance, fairness chart and a plain-language explanation.

- **Backend:** Python 3.11, FastAPI, Pydantic, SQLite.
- **ML:** scikit-learn, XGBoost, SHAP, pandas, joblib.
- **Frontend:** React, Vite, TypeScript, Tailwind CSS, shadcn/ui, Recharts, Framer Motion; light and dark mode; INR formatting.

Roadmap and Value

Value today	Next	Later
Faster, more consistent decisions; explanations officers can share; earlier warning on portfolio risk.	Authentication and roles, audit logs, bureau data integration, model monitoring and drift alerts.	Policy simulation, more loan products, regulator-ready reporting and multi-currency support.

Limitations to be honest about

- Prototype trained on public or synthetic data; it must be revalidated on a bank's own portfolio before any real use.
- No authentication or paid data sources in this version.

CreditSense

Clearer credit decisions. Earlier risk signals.

Faster for officers. Fairer and more explainable for applicants. Safer for the portfolio.

Thank you